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Incipient ferroelectrics: Anomalous T_1 behaviors and their rotor interpretation

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ABSTRACT

The quantum temperature (denoted by T_1) behaviors of three typical incipient ferroelectrics, $SrTiO_3$, $KTaO_3$ and $CaTiO_3$, are studied. This quantity is argued to serve fundamentally in identifying the nature of the local mode responsible for the dielectric responses. Our main findings are as follows. For all compounds, T_1 saturates at low temperatures. For $CaTiO_3$, T_1 monotonically increases with temperature and no clear saturation is discernible at high temperatures. For $KTaO_3$, similar behaviors are observed but with a little twist: a dip shows up around 35 K, above which T_1 increases but below it T_1 decreases with temperature. Although it is hardly seeable in this compound, this dip might mark a transition, whose nature is unclear for the moment. In parallel with $KTiO_3$, $SrTiO_3$ also has a dip, which is much stronger and broader. It happens around 105 K, at which the famous anti-ferrodistortive (AFD) transition occurs. Were it not for this dip, T_1 would drop to zero in $SrTiO_3$ at low temperatures and the ferroelectric (FE) transition would take place. The dip halts the drop and makes T_1 rise up to a value that is enough to stabilize the FE instability. In this respect, the dip is essential in preventing the FE transition in $SrTiO_3$. Since the dip and the AFD transition occur at roughly the same temperature, we attempt to ascribe the former to the latter. This ascription is compatible with previous work [A. Yamanaka, M. Kataoka, Y. Inaba, K. Inoue, B. Hehlen, E. Courtens, Europhys. Lett., 50:(2000) 688]. To interpret the T_1 behaviors, we utilize an anisotropic rotor model, according to which the local variable is supposed to move on a non-uniform sphere. By tuning the anisotropy parameter, υ, qualitative agreement can be achieved. Especially, a single $\upsilon \approx 100$ can fit the T_1 of $CaTiO_3$ over the entire temperature range under consideration, whereas the fitting for $KTaO_3$ requires two different υ, namely, $\upsilon \approx 260$ above the dip temperature and $\upsilon \approx 40$ below it. Analogously, two υ are also required for $SrTiO_3$. Below the dip temperature, a very good fitting can be obtained with $\upsilon \approx 40$. We did not try to fit the high temperature data of $SrTiO_3$, because the data in this range are scarce and inaccurate. Nevertheless, we believe that a different and bigger υ should be at work, considering the case with $KTaO_3$. Assuming the AFD transition as the cause of the dip in $SrTiO_3$, we may claim that, the true role of the AFD transition in stabilizing the FE instability is to reduce the υ and then enhance quantum fluctuations.

1. Introduction

Recent years have seen revived interest in incipient ferroelectrics (FE), such as $SrTiO_3$ (ST) and $KTaO_3$ (KT). These materials display huge dielectric constants as the temperature T goes down, but never reach a FE phase transition. FE fluctuations and quantum fluctuations (QF) are believed to compete in them [1,2]. The former is usually characterized by a temperature T_0 while the latter by T_1. The competition is best illustrated in isotope-exchanged ST, $SrTi(O^{16}_{1-x}O^{18}_{x})_3$, where $T_1 - 2T_0$ evolves gradually from a positive value at small $x < x_c$ to a negative value at big $x > x_c$ [1]. Here x_c marks a quantum critical point. A flurry of work has been devoted to exploring the criticality [3–6].

Perhaps the simplest formula that captures the competition phenomena was derived by Barrett [7]. It relates the dielectric constant ϵ to T by $\epsilon_{bar} = \epsilon_\infty + \frac{A}{\frac{T_1}{2}\coth(\frac{T_1}{2T})-T_0}$. In the high T limit, the classical result is restored and $\epsilon_{bar} \sim \frac{A}{T-T_0}$, while in the low T limit, it saturates as $\epsilon_{bar} \sim \frac{2A}{T_1-2T_0}$. Both features agree with the experimental data of ST and KT [8,9], thus making the formula a smoke-gun for identifying new incipient FE such as $CaTiO_3$ (CT) [10]. Nonetheless, systematic deviations at intermediate temperatures from ϵ_{bar} were noticed almost back to three decades ago when ST was first claimed as an incipient FE [8]. Such deviations were somewhat remedied by invoking a T-dependent T_1, which was assumed to transit from a low T value T_{1l} to a high T value T_{1h} [8,11,12]. In ST, this issue is further complicated by the anti-ferrodistortive (AFD) transition at $T_a \approx 105$ K. This transition is non-polar, but recent work [13] has demonstrated its crucial role in preventing the FE in ST. Specifically, were it not for this

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transition, ST would undergo a polar transition around 30 K. In what follows, we disclose unexpected behaviors of T_1, which are argued to be subsequent to the AFD transition.

We notice several reasons to focus on T_1 instead of ϵ. Firstly, the behaviors of T_1 are much more revealing than those of ϵ. Apparent agreements with ϵ can be easily achieved, almost regardless of how the T_1 behaviors are postulated [8,11,12]. Secondly, T_1 may be regarded as an independent quantity with clear physical meanings, rather than a simple fitting parameter. From a mean-field point of view, ϵ hinges on temperature primarily through the bare electric susceptibility $\tilde{\chi}_0$, which is associated with motions of the local mode $\vec{u}$ belonging to a single unit cell and proportional to the electric dipole as defined in Ref. [14]. Different modeling of $\vec{u}$ leads to different $\chi_0(T)$. Often, one may approximate the behaviors of $\vec{u}$ via a bistable system that is characterized by a frequency $\frac{k_B T_1}{\hbar}$, at which the system rustles. In pseudo-spin modeling, T_1 is simply a constant and measures how fast the spin flips, whereas in double-well modeling (DWP [15]), T_1 decreases with decreasing temperature and eventually saturates at low temperatures [16]. In such cases, one easily derives the Barrett formula in relating T_1 to $\tilde{\chi}_0$. However, a broader meaning may be assigned by noting that T_1 actually indicates the low temperature stiffness of the system, because, assuming the validity of the Barrett formula, $\tilde{\chi}_0^{-1} \propto T_1$ for $T/T_1 \ll 1$. Thirdly, as already mentioned, the behaviors of T_1 are model dependent, implying they can serve as fingerprints in identifying the nature of the local mode.

In the present work, we extract the empirical T-dependence of T_1 for three typical incipient FEs, ST, KT and CT. The experimental data have been well documented on ϵ and the following equation will be solved to find T_1 for each T,

$$\frac{T_1}{2}\coth\left(\frac{T_1}{2T}\right) = T_0 + \frac{A}{\epsilon - \epsilon_\infty}. \qquad (1)$$

Here T_0 and A^{-1} are found as the extrapolated Curie temperature and the slope of the straight segment on the $\frac{1}{\epsilon-\epsilon_\infty}$ vs. T curve, respectively. Our main results are the following:

(1) In CT, T_1 starts with a nearly flat section at low T, followed by a linear section, where T_1 increases monotonically without any discernible saturation at high T. This is qualitatively compatible with the DWP model. But quantitatively, we *cannot* fit the behavior by this model. No DWP parameter could be found to fit both the flat and the increasing part simultaneously.

(2) In ST, T_1 does not simply increase with temperature. Rather, it at first diminishes in a very smooth way down to zero around the AFD critical temperature T_a, and then starts to rise up. Up to our knowledge, this dip behavior has never been reported before. And it *cannot* be consistent with the DWP at all. Likely, it originates from the AFD rotation.

(3) In KT, a similar dip behavior is noticed. But here the dip is very weak and narrow and almost imperceptible. Its nature is unidentified.

(4) The T_1 behaviors in all the materials under consideration can be understood using a $O(3)$ rotor model in the presence of some cubic intra-cell anisotropy (υ). We shall call it the anisotropic rotor model (ARM). It turns out that, two different υ are required for ST and KT, corresponding to the low-T and high-T section, respectively.

The T_1 behaviors obviously go beyond the DWP description. The remainder of this paper is organized as follows. In the next section, the extraction of T_1 versus T is detailed. The ARM is described in the third section, where the fitting using this model is done. A summary is given in section four.

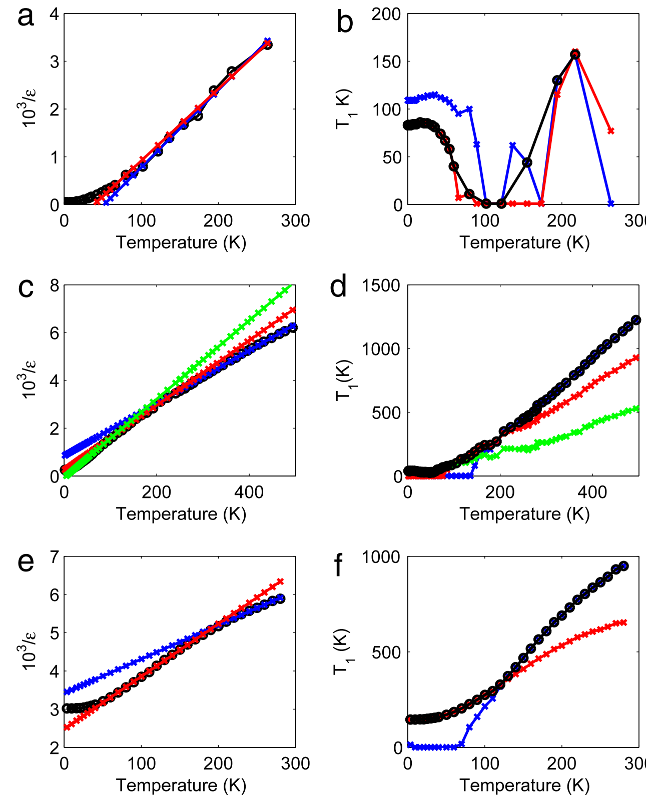

Fig. 1. The experimental inverse dielectric constant, $1/\epsilon$ and the extracted T_1 by the method described in the text. (a) and (b): ST, ϵ after Ref. [8]; (c) and (d): KT, ϵ after Ref. [9]; (e) and (f): CT, ϵ after Ref. [10]. The black curves in the T_1 panels are obtained by grafting relevant segments from the colored curves, as described in the text.

2. Extraction of T_1 using available experimental data

The general procedures to evaluate T_1 from the experimental ϵ and ϵ_∞ have been outlined in the introduction. However, there is a certain subtleness: the high T section of the experimental $\frac{1}{\epsilon}$ seems broken into two or three straight segments, as seen in Fig. 1. Thus, slight variations may have occurred to A and T_0, which can be caused by changes in e.g. the couplings between the local modes belonging to different unit cells. Hence, the temperature dependence of $\frac{1}{\epsilon}$ may be due to (A, T_0) or T_1 or both. One must isolate the latter (as due to T_1) to pursue genuine behaviors of T_1. This is done by using segment-wise (A, T_0), which are obtained by the usual least-square fit method, to deduce T_1. The final T_1 is obtained by grafting the segments together.

The results are exhibited in Fig. 1. As we see, for ST, two segments are applied, as indicated by the red and blue curves in Fig. 1(a) and (b), where the black curve stands for the result after grafting. Similar annotations are given to other panels in Fig. 1. In comparison with CT and KT, what is special about ST is the huge dip of T_1 around $T_a \sim 105$ K. It is natural to ascribe the dip to the AFD transition, which occurs at the same temperature. This ascription is consistent with other work [12,13]. In Ref. [13], the authors performed high-resolution hyper-Raman scattering (HRS) measurements on ST, by which the temperature dependences of soft polar mode frequency, ω_0, were determined. Although above T_a the ω_0^2 follows perfectly the Curie–Weiss law, it starts to split into two branches (at ω_a and ω_c, respectively) and deviate from this law significantly after crossing T_a. Then they wrote $\omega_{a,c}^2$ as a sum of two terms, the first supposed to be the squared frequencies that would result if the AFD did not take place and the second being proportional to the square of the AFD order parameter. After determining the second term independently via other methods, they were able to settle the first one, which actually softens to zero at around 30 K, indicating FE instability. However, this instability was eventually prevented by the AFD rotation. They concluded that, QF alone could not suppress the FE, thus confirming the essential role of AFD. Now, a meticulous inspection of Fig. 4 in Ref. [13] reveals more: the low T parts of ω_0^2 and $\omega_{a,c}^2$ take analogous shapes that are characteristic of QF, the only distinction being that the latter implies stronger QF. Hence, it is safe to assert that, the AFD acts to enhance QF. In the next section, this assertion is revisited and the enhancement channel is unveiled.

In closing this section we mention that, there is also a similar dip around 35 K in KT, where three segments were used. But this dip is far less prominent than that in ST. Although it is hardly seeable in the grafted curve, it has important consequences, as we will argue in what follows. This is unexpected, since no structural instabilities in KT around 35 K seem to exist.

3. Fitting T_1 by ARM

3.1. Brief derivation of ARM

This subsection is devoted to the derivation of ARM. According to Refs. [14,17], $V(\vec{u})$ for all the materials under interest can be couched as $V(\vec{u}) = \alpha u^4 - \kappa u^2 - \gamma(u_x^2u_y^2 + u_y^2u_z^2 + u_z^2u_x^2)$. Here u is the magnitude of $\vec{u} = (u_x, u_y, u_z)$. All coefficients are assumed positive and satisfy the stability condition $\gamma < 3\alpha$. It is easy to show that, V has eight global minima situated at the corners, six local minima pointing to the face centers, twelve pointing to the edge centers and a maximum at the center of a cube, whose body diagonal has a length that is twice $l_0 = \sqrt{\frac{3\kappa}{6\alpha-2\gamma}}$. Obviously, for huge l_0, the low energy description of the model can be approximated by the so-called eight-site model (8SM), which has been the prototypical model in studying the order–disorder phenomena in ferroelectrics [18]. In this extreme, the relevant degrees of freedom are the orientations instead of the magnitude of $\vec{u}$. This motivates us to work with a polar coordinate frame, in which $\vec{u} = u(n_x = \sin\theta\cos\varphi, n_y = \sin\theta\sin\varphi, n_z = \cos\theta)$, $V = [\alpha - \gamma f(\theta,\varphi)]u^4 - \kappa u^2$ and $\nabla^2 = \frac{1}{u^2}\frac{\partial}{\partial u}(u^2\frac{\partial}{\partial u}) - \frac{\hbar^2\hat{\Omega}^2}{u^2}$, where $\hbar\hat{\Omega}$ is the angular momentum and $f(\theta,\varphi) = \frac{1}{2}[1-(n_x^4+n_y^4+n_z^4)]$.

We see that the V features a trough around $l(\theta,\varphi) = \sqrt{\frac{\kappa}{2[\alpha-\gamma f(\theta,\varphi)]}}$ in any direction. Given $\vec{n}$, there is only one trough, because u is always positive by definition. In this sense, the radial motion can somewhat be 'frozen in' at relatively low temperatures due to the trough. Therefore, we can average it out to get a model including explicitly only angular variables. Up to a trivial constant, one arrives at

$$\tilde{H}_{\mathrm{ARM}} = \frac{\hbar^2\,\hat{\Omega}^2}{2I} + \tilde{\upsilon}(n_x^4 + n_y^4 + n_z^4);$$
$$\frac{1}{u_0^2} = \left\langle\frac{1}{u^2}\right\rangle, \qquad I = Mu_0^2, \qquad \tilde{\upsilon} = \frac{\gamma}{2}\langle u^4\rangle \tag{2}$$

which is just the ARM. There is an intrinsic energy scale $\varepsilon_0 = \frac{\hbar^2}{2I}$, by which we can define a dimensionless constant $\upsilon = \tilde{\upsilon}/\varepsilon_0$. Reasonably, we expect that $\langle u\rangle \approx u_0 \approx l_0$.

3.2. Computing the susceptibility of ARM

To do the fitting, we have to calculate the susceptibility $\tilde{\chi}_0$. Let Q be the Born effective charge associated with the local mode $\vec{u}$. The dipole moment is then given by $\vec{P} = Q\vec{u}$, which is coupled to an external electric field $\vec{E} = (0, 0, \tilde{E})$ via $-P_z\tilde{E}$. Then we have $\tilde{\chi}_0 = \frac{\partial\langle P_z\rangle}{\partial\tilde{E}}$. For convenience, we also introduce a dimensionless susceptibility χ_0 so that $\tilde{\chi}_0 = \frac{Q^2\langle u\rangle^2}{a^3\varepsilon_0}\chi_0$, where a denotes the lattice constant. It is easy to show that, $\chi_0 = \frac{\langle n_z\rangle}{\partial E}$. Here $E = Q\langle u\rangle\tilde{E}/\varepsilon_0$. Now expressing energies in unit of ε_0, we obtain a dimensionless Hamiltonian

$$H_{\mathrm{ARM}} = \frac{\hat{\Omega}^2}{2} + \upsilon(n_x^4 + n_y^4 + n_z^4) - En_z,$$
$$H_{\mathrm{ARM}} = \frac{\tilde{H}_{\mathrm{ARM}}}{\varepsilon_0}. \tag{3}$$

Here the external perturbation has been added explicitly. Note that $\langle n_z\rangle = -\langle\frac{\partial H_{\mathrm{ARM}}}{\partial E}\rangle$. Let $\epsilon_N(E)$ be the N-th eigenvalue of H_{ARM}, then $\langle n_z\rangle = -\sum_N \rho_N\frac{\partial\varepsilon_N(E)}{\partial E}$, where $\rho_N = \frac{\exp(-\frac{\varepsilon_N}{t})}{\sum_{N'}\exp(-\frac{\varepsilon_{N'}}{t})}$ is the Boltzmann factor. Similarly, $\chi_0 = -\sum_N \rho_N\frac{\partial^2\varepsilon_N(E)}{\partial E^2}$. Here all the derivatives are taken at $E = 0$ and $t = \frac{k_BT}{\varepsilon_0}$ is the reduced temperature.

Now we need to find all the ε_N by solving the corresponding secular equation. This can be done only numerically. The representation will be chosen to be the spherical harmonics $|l, m\rangle$, namely, $\langle\theta,\varphi|l, m\rangle = Y_l^m(\theta,\varphi)$. In practice, the Hilbert space must be truncated. Currently, we truncate it at $L_c = 12$. Thus, in total L_c^2 eigenvalues will be obtained. $\hat{\Omega}^2$ is diagonal in this representation, $\langle l', m'|\hat{\Omega}^2|l, m\rangle = \delta_{l,l'}\delta_{m,m'}l(l+1)$. To compute the matrix elements of $n_{x,y,z}^4$ and $n_{x,y,z}$, we express all these quantities in terms of Y_l^m. It is easy to check that

$$n_x = \sqrt{\frac{2\pi}{3}}(Y_1^{-1} - Y_1^1), \qquad n_y = i\sqrt{\frac{2\pi}{3}}(Y_1^{-1} + Y_1^1)$$
$$n_z = \sqrt{\frac{4\pi}{3}}Y_1^0 \tag{4}$$
$$n_x^4 + n_y^4 + n_z^4 = \frac{\sqrt{\pi}}{1260}(Y_4^4 + Y_4^{-4}) + \frac{4\sqrt{\pi}Y_4^0}{15} + \frac{6\sqrt{\pi}Y_0^0}{5}.$$

Now the required matrix elements can all be built with the following quantity

$$\int(Y_{l_1}^{m_1})^*\cdot Y_{l_2}^{m_2}\cdot Y_{l_3}^{m_3} = (-1)^{m_1}\int Y_{l_1}^{-m_1}\cdot Y_{l_2}^{m_2}\cdot Y_{l_3}^{m_3}$$
$$= (-1)^{m_1}C_{l_1l_2l_3}\times W_{0,0,0}^{l_1,l_2,l_3}\times W_{m_1,m_2,m_3}^{l_1,l_2,l_3}$$
$$C_{l_1l_2l_3} = \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}}. \tag{5}$$

Here the integral is taken over the solid angle, the first line ensues from the property of spherical harmonics and in obtaining the second line we have used Wigner 3-j symbols W_m^l [19]. Now plug the matrix forms of all quantities into H_{ARM} and the eigenvalues can be found in a standard way. Then the χ_0 can be computed with the prescription offered above.

3.3. ARM vs. experimental

To make comparison with the T_1 prescribed in Section 2, we need to establish the relation between T_1 and χ_0. From a mean-field point of view, one should have $\tilde{\chi}_0 = \frac{Q^2\langle u\rangle^2}{a^3\varepsilon_0}\cdot\frac{1}{\chi_0^{-1}-\upsilon}$, where υ is a dimensionless constant representing the molecular field strength that pertains to T_0. Then the dielectric constant can be written as $\epsilon = \epsilon_\infty + \frac{\chi_0}{\epsilon_0} \equiv \epsilon_\infty + \frac{\tilde{A}}{\chi_0^{-1}-\upsilon}$, where ϵ_0 denotes the vacuum permittivity and $\tilde{A} = \frac{Q^2\langle u\rangle^2}{a^3\varepsilon_0\epsilon_0}$. Now we expect that, to have the Curie–Weiss law, χ_0^{-1} should follow a simple linear relation

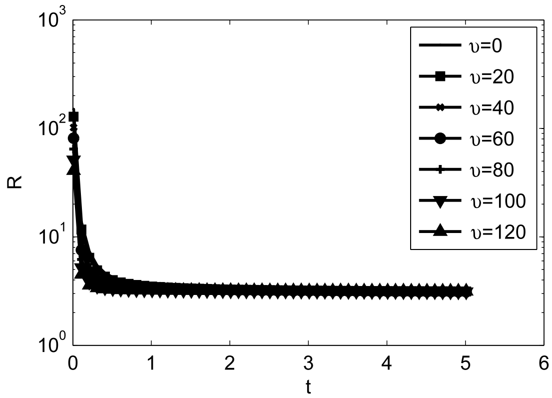

Fig. 2. The behaviors of R vs. t at various υ. All curves fall on top of each other at large t, indicating the universality of the high t limit of R.

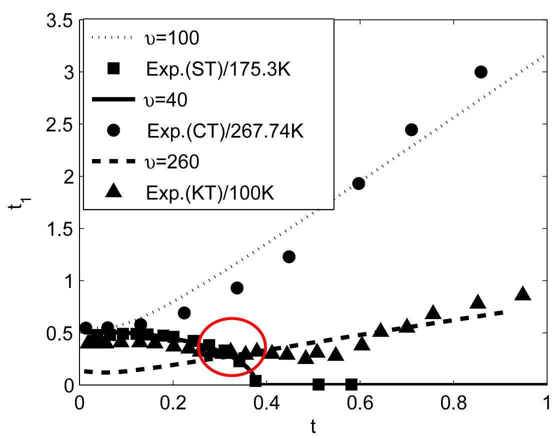

Fig. 3. Comparison between the theoretical and experimental t_1 against t, with $R^* = R(t = 3000)$. The agreement is much better for ST and CT than for KT. In KT, as indicated by the red circle, there seems to be a transition around 35 K, where a mild dip is observable. But to our knowledge, there seem not to exist other accompanying transition-like behaviors such as AFD in this compound. Note that the dotted curve (with $\upsilon = 260$) fits KT qualitatively only beyond the dip, while the low temperature part is well fit with $\upsilon = 40$ and $\frac{\varepsilon_0}{k_B} \approx 150$ K. We did not try to fit the high-t part of the ST curve, because the experimental data seems not accurate enough.

with t, namely, $\chi_0^{-1} \overset{\text{for large } t}{\longrightarrow} R(t) \cdot t$, with R approaching a constant at high t. Numerically, we have computed R versus t at various υ and found that, R approaches a seemingly universal constant $R^* \approx 3$ regardless of υ, as illustrated in Fig. 2. Making use of this knowledge, we may define T_1 with χ_0 by the following equation

$$\frac{1}{R^*\chi_0} = \frac{t_1}{2}\coth\left(\frac{t_1}{2t}\right), \qquad T_1 = t_1 \times \frac{\varepsilon_0}{k_B} \tag{6}$$

which can be solved numerically for every t [20] and then the as-found T_1 can be compared with that extracted experimentally. This definition is compatible with the way of extracting T_0 and A in Section 2. In the present jargon, one recognizes that $A = \tilde{A}/R^*$ and $T_0 = (\upsilon/R^*) \times \frac{\varepsilon_0}{k_B}$.

The comparison is summarized in Fig. 3. The $\varepsilon_0/k_B \approx 175$, 267 and 100 K have been chosen for ST, CT and KT, respectively [21]. A single $\upsilon = 100$ gives a rough fit for CT over the entire temperature range under consideration. For KT, however, two different υ are required, namely, $\upsilon \approx 260$ above the dip temperature and $\upsilon \approx 40$ below it. Analogously, two υ are also required for ST, in which the same $\upsilon \approx 40$ can fit the data below the dip. We did not try to fit the high temperature data of ST, because the data in this range are scarce and inaccurate. Nevertheless, we believe that a different and bigger υ should be at work, considering the case with KT. Obviously, smaller υ means larger QF and vice versa. Therefore, QF in ST and KT is stronger at low t than at high t. This is consistent with the conclusion drawn in Ref. [13], as emphasized previously.

4. Summary

In summary, we have analyzed the T_1 behaviors of three typical incipient FEs, both experimentally and theoretically. Unexpected phenomena have been revealed. Although in all three compounds the T_1 generally increases with temperature, strange behaviors appear as they are cooled down. In ST, a dip is found around 105 K. This is deciphered as a sequela of the AFD transition that occurs at the same temperature. In KT, a similar but much more moderate dip is also seen, around 35 K, at which, however, no structural transition seems to take place. In CT, no such dip is found. To understand such unusual behaviors, we have constructed an anisotropic rotor model, which reduces to the eight-site model at huge anisotropy. It turns out that, qualitative agreement can be achieved between the experimental observations and the ARM calculations for these compounds. In particular, to fit ST at low temperatures, a very small υ (40) has been used, while at high temperatures, we believe υ should be much bigger. In this sense, quantum fluctuations are stronger at low than at high temperatures. This accords with other work. It is safe to say that the true role of AFD in preventing FE in ST is to enhance quantum fluctuations by diminishing the anisotropy υ.

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[19] Eric W. Weisstein, Wigner 3j-Symbol, From MathWorld (http://mathworld.wolfram.com/Wigner3j-Symbol.html).
[20] Note that the r.h.s of this equation is an even function of and increases with t_1 given t. It has a single minimum of t at $t_1 = 0$. Therefore, if the l.h.s is less than t, no real solution will exist. On such occasions, we always choose $t_1 = 0$. This trick also applies in solving Eq. (1).
[21] These ε_0 correspond to u_0 of the order of 10 pm, probably larger than what is expectd of reality. But the order is basically correct.